

Name	Subject Class	Class	Candidate Number
	2BI		



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2017

BIOLOGY

HIGHER 2

9744/02
17 AUGUST 2017
2 hours

Paper 2

READ THESE INSTRUCTIONS FIRST

Write your name, index number and class on this answer booklet.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
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8	
9	
Total	100

This question paper consists of **22** printed pages.

1 Fig. 1.1 shows part of a cell.

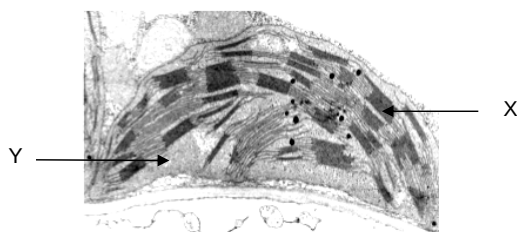


Fig. 1.1

(a) (i) Outline the role of the organelle labelled X.

1. **Contain photosystem/ pigments for the synthesis of ATP and reduced NADP via light dependent reaction/ photophosphorylation;**
2. **For carbon fixation/ synthesis of glucose/ GALP via the Calvin cycle/ light-independent reaction;**

[2]

(ii) Explain the importance of the double membrane enclosing organelle X for the reactions that occur in the region labelled Y.

1. **Increases the concentration of enzyme and/or substrate for higher rate of Calvin cycle;**
2. **Maintain optimum conditions for the enzymes in Calvin cycle;**
3. **Separate reactions of the Calvin cycle from incompatible reactions (in the cytoplasm like glycolysis);**

R! reference to thylakoid membrane for production of H^+ pool

[2]

(b) (i) Plant cells are unable to carry out endocytosis due to the energy needed to overcome the turgor pressure in the plant cell. However, lysosome-like organelles can be found in the plant cells. Suggest reasons for their presence in plant cells.

1. **Digest/ carry out autophagy of old and worn out organelles;**
2. **Break down macromolecules (by hydrolysis)/ digest ingested particles;**
3. **Autolysis/ apoptosis of cell;**

[2]

(ii) Outline how the lysosome is formed by the endomembrane system in a eukaryotic cell.

1. **Transcription of genes coding for lysosomal proteins in the nucleus;**
2. **Synthesis of lysosomal proteins by ribosomes via translation on the RER;**
3. **Folding/post-translational chemical modification in the RER;**
4. **Transport of lysosomal proteins via transport vesicles from RER to GA**
5. **Post-translational chemical modification/modification, sorting and packaging of lysosomal proteins in the GA;**
6. **Vesicles containing lysosomal proteins pinches off GA to form lysosome;**

[3]

The chloroplast is a member of a larger family of plant organelles called plastids. All plastids, including chloroplasts, develop from proplastids. Fig. 1.2 shows the process of chloroplast development from a proplastid.

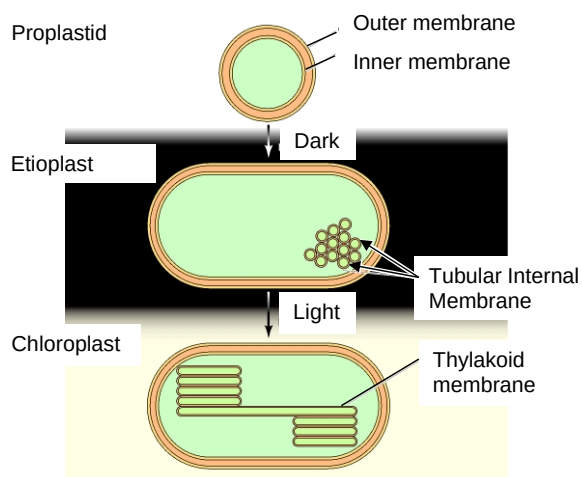


Fig. 1.2

(c) With reference to Fig. 1.2, suggest how thylakoid membranes are formed.

1. * **Pinching / invagination + fusion of the inner membrane / budding of the inner membrane of proplastids;**
 2. **Stimulated by transition from light to dark to form a tubular internal membrane / membrane tubules / vesicle;;**
 3. **Light stimulates the tubules to form thylakoid membrane;**
 4. **Elongation of tubules to form thylakoid membranes;**
- (Max. 1 from pts 1-2)

[2]

[Total: 11 m]

2 Fig. 2.1 shows the formation of a bond in the synthesis of starch.

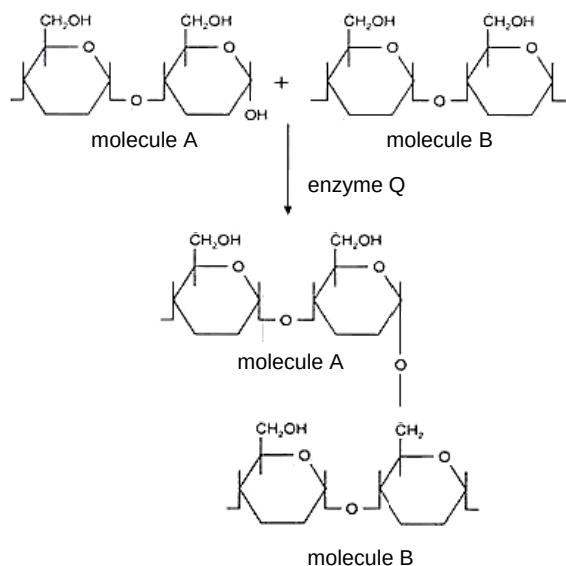


Fig. 2.1

(a) (i) Describe the formation of the bond in Fig. 2.1.

1. α -1,6-glycosidic bond formed;
2. Between -OH groups of C1 of a glucose monomer in molecule A and C6 of a glucose monomer in molecule B;
3. Release of a water molecule via condensation reaction;
4. Catalysed by an enzyme Q;

[2]

(ii) Explain how enzyme Q could lower the activation energy of the bond formation.

1. Enzyme Q has an active site which is complementary/ specific to the substrates molecules A and B;
2. Resulting in the formation of an enzyme-substrate complex;
3. Orientates molecules A and B accurately for bond formation/ Increases proximity of molecules A and B for bond formation/ Active site provides suitable environment for condensation reaction/ distortion of bonds within molecule A and B;
4. Reactants require less energy to reach transition state;

[3]

Graph X shows the amount of the product formed over time for an enzyme-catalysed reaction (Fig. 2.2).

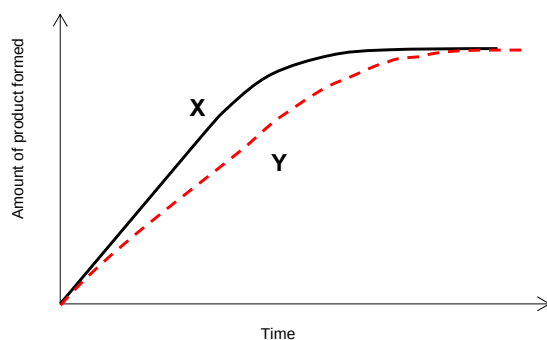


Fig. 2.2

(b) (i) Draw a graph, labelled Y, on Fig. 2.2 to show how a small amount of non-competitive inhibitor affects the amount of product formed over time. [1]

(ii) Explain the difference between graphs X and Y.

1. Graph Y has a gentler gradient/ Lower rate of product formation / takes longer for final amount of product to be formed
2. Less free enzymes available as inhibitor binds to a site other than the active site (R! allosteric site);
3. Causing 3D conformation of enzyme to change, and enzyme active site is no longer complementary to the substrate; [2]

[Total: 8m]

- 3 The diagram below shows the structure of a mature tRNA for the amino acid alanine.

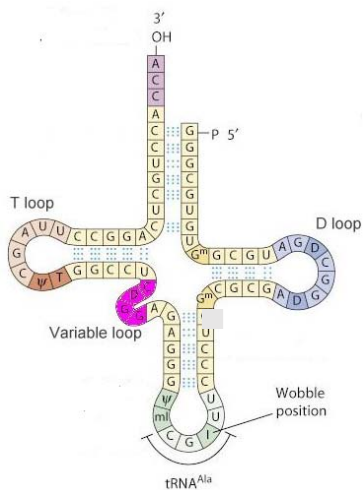


Fig. 3.1

- (a) (i) Explain the roles of hydrogen bonds in the proper functioning of tRNA.

1. Hydrogen bonds are formed between complementary base pairs allows folding of tRNA into (specific) 3D conformation / looped structures / clover-shape;
2. For stability (during activation and translation);
3. To fit into the complementary binding site of ribosome (P/A site) / active site of aminoacyl-tRNA synthase;
4. Hydrogen bonds form between anticodons on tRNA with complementary codons on mRNA;
5. To translate codon sequence into amino acid sequence;

[4]

Note: mark once for concept of complementary base pairing

- (ii) The length of the tRNA gene is longer than that of the mature tRNA. Outline how a tRNA molecule is synthesised in eukaryotes.

1. RNA polymerase binds to promoter of tRNA gene to transcribe tRNA;
2. Introns are excised and exons are spliced together;
3. Folding of tRNA into clover shape via the formation of H bonds between complementary base pairs;
4. Addition of 3'CCA sequence by enzymes / addition and removal of 5' cap; (additional info)

[3]

For
examiner's
use

An experiment was carried out to investigate the effects of various cytokines on a culture of CD34 cells. CD34 cells are hematopoietic progenitor cells which are produced in the early stage of differentiation to form mature immune cells from stem cells. Three types of cytokines, KF36EG, K36EG and F36EG, were used in the experiment as shown in Fig. 3.2.

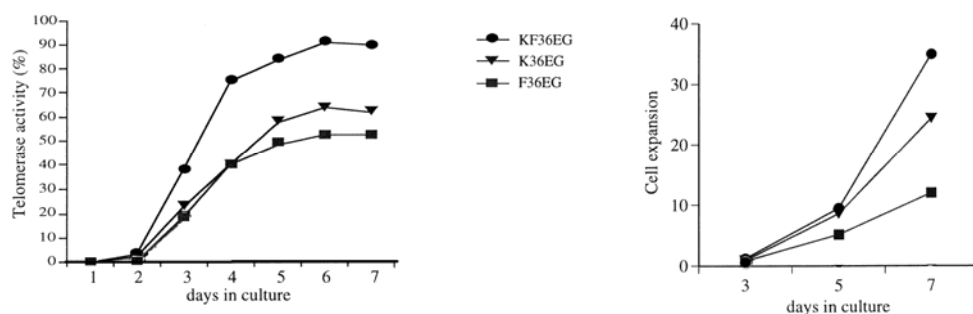


Fig. 3.2

(b) (i) With reference to Fig. 3.2, describe the effects of the types of cytokines on CD34 cells.

- * All three cytokines led to an increase in telomerase activities (from day 0 to day 7) + All three cytokines led to an increase in cell expansion from day 3 to day 7;
- KF36EG caused the highest increase in telomerase activities as compared to K36EG and F36EG + KF36EG caused the highest rate of cell expansion at 35 a.u. as compared to K36EG (24 a.u) and F36EG (11 a.u);
- Reference to data with comparison; Refer to data (max. 1 m)
Telomerase activity increases from 3% to 90% from day 2 to day 7 for KF36EG
+ Telomerase activity increases from 3% to 60% from day 2 to day 7 for K36EG / Telomerase activity increases from 3% to 52% from day 2 to day 7 for F36EG;
- Rate of increase of telomerase activities decreased from day 4 to day 7;

[3]

* Compulsory point

(ii) With reference to Fig. 3.2, comment on the role of cytokines on the CD34 cells.

- Cytokine is a signalling molecule;
- That promotes mitosis/proliferation/cell expansion (and differentiation) of CD34 cells;
- By increasing the rate of transcription / activity of telomerase in CD34 cells;
- Allows CD34 cells to lengthen telomeres for subsequent rounds of DNA replication;

[2]

(c) Discuss the accuracy of the following statement:

"Stem cells and cancer cells are able to divide indefinitely as there is no end replication problem due to the presence of active telomerase."

1. Stem cell and cancer cells are able to divide indefinitely due to the presence of active telomerase is accurate;
 2. So as to prevent triggering of replicative senescence as length of telomeres is maintained;
 3. However, there is end replication problem in both cells;
 4. As DNA polymerase cannot synthesise a complementary DNA strand from scratch / DNA polymerase requires a primer / an free 3' OH end of an existing strand to synthesise a complementary strand;
 5. Hence the 3' end of the leading strand will not be fully replicated (forming an overhang);
 6. Active telomerase can only lengthen the 3' end for DNA replication to prevent erosion of genes due to end replication problem;
 7. Cells would also require other factors such as ATP and nutrients for replication / AVP;
- [4]

[Total: 16 m]

- 4 The coat colour of Norwegian cattle is mainly determined by the distribution of two pigments: red and black. Both pigments are produced by the action of the enzyme tyrosinase in cells called melanocytes. Low enzyme activity leads to the production of red pigment, while high enzyme activity brings about black pigment production.

The activity of the enzyme is increased when melanocyte stimulating hormone (MSH) combines with a MSH receptor. The receptor is coded for by the gene, **R**, which has three alleles, **R^D**, **R^A** and **r**. **R^D** and **R^A** each codes for a receptor with a different activity. No receptor is produced by the recessive allele, **r**.

The dominant allele of a second gene, **B**, codes for a protein which binds to and blocks the MSH receptors coded for by **R^A**, thus preventing stimulation of tyrosinase activity in a melanocyte. The receptors coded for by **R^D** is insensitive to the protein coded by **B**. The recessive allele, **b**, does not produce a functional protein.

- (a) (i) State the name given to the interaction between the **R** and **B** gene loci.

Epistasis; [1]

- (ii) Explain why animals with the genotype **R^AR^ABB** have red coats.

Inhibitor / Protein that blocks MSH receptor coded for/produced and hence MSH unable to bind to its receptor to increase/stimulate tyrosinase / enzyme activity; [1]

- (iii) A red cow, with genotype **R^AR^ABB** is mated with a bull which is homozygous recessive at both gene loci.

Draw a genetic diagram in the space below to show the expected genotypes and phenotypes and their ratios in the F₁ and F₂ generations.

Parent phenotypes	Red bull	x	Red cow
Parent genotypes	rrbb		R ^A R ^A BB
Gametes	(rb)		(R ^A B)
F ₁ genotype	R ^A rBb		
F ₁ phenotype	All red		
F ₁ cross	R ^A rBb	x	R ^A rBb
F ₁ gametes	(R ^A B) (R ^A b)		(rB) (rb)



Punnett Square:

	$\text{R}^{\text{A}}\text{B}$	$\text{R}^{\text{A}}\text{b}$	rB	rb
$\text{R}^{\text{A}}\text{B}$	$\text{R}^{\text{A}}\text{R}^{\text{A}}\text{BB}$ Red	$\text{R}^{\text{A}}\text{R}^{\text{A}}\text{Bb}$ Red	$\text{R}^{\text{A}}\text{rBb}$ Red	$\text{R}^{\text{A}}\text{rBb}$ Red
$\text{R}^{\text{A}}\text{b}$	$\text{R}^{\text{A}}\text{R}^{\text{A}}\text{Bb}$ Red	$\text{R}^{\text{A}}\text{R}^{\text{A}}\text{bb}$ Black	$\text{R}^{\text{A}}\text{rBb}$ Red	$\text{R}^{\text{A}}\text{rb b}$ Black
rB	$\text{R}^{\text{A}}\text{rBB}$ Red	$\text{R}^{\text{A}}\text{rBb}$ Red	rrBB Red	rrBb Red
rb	$\text{R}^{\text{A}}\text{rBb}$ Red	$\text{R}^{\text{A}}\text{rb b}$ Black	rrBb Red	rrbb Red

F₂ phenotypic ratio **13 red coat : 3 black coat**

[5]

Mark allocation:

1. Correct parent gametes;
2. Correct F₁ phenotype and genotype;
3. Correct F₁ gametes;
4. Correct F₂ phenotypes and genotypes in Punnett square;
5. Correct F₂ phenotypic ratio;

During a health screening exercise of cattle in a farm, the height of the bulls was measured and the data collected is shown in Table 4.1.

Table 4.1

Height/cm	Number of bulls
131—135	3
136—140	9
141—145	21
146—150	12
151—155	2

- (b) Distinguish between the two types of variation shown in coat colour and height in the Norwegian cattle.

1. * Coat colour shows discontinuous variation	While height shows continuous variation;
2. Discrete phenotypic classes and no intermediates are observed	A range of phenotypes are observed;
3. Coat colour is controlled by one or two major genes, which may have two or more allelic forms.	Height is controlled by a large number of genes (polygenes);
4. Effect of individual genes can be observed.	Effect of individual genes cannot be observed;
5. Effect of genes is not additive.	Effect of genes is additive;
6. The environment has a small effect on the phenotype.	Environment has a large effect on the phenotype;

* Compulsory point;

[3]

In the honey bee colony, the queen bee is solely responsible for laying eggs and the drones for fertilizing her. The worker bees have well-developed mouthparts and structural adaptations for collecting nectar and pollen to gather food and to perform other duties in the hive. Male bees are developed from haploid eggs while both queen and worker bees develop from fertilized eggs.

- (c) Explain how the phenotypic differences between the queen and the worker bees come about despite both being developed from fertilized eggs.

1. The different diets brought about differences in gene expression in cells, resulting in the phenotypic differences;
2. Larvae fed on diet of royal jelly throughout development become queen bees while those fed on worker jelly become worker bees;

[2]

[Total: 12 m]

- 5 Normal cells rely on oxidative phosphorylation in the mitochondria to generate the energy needed for cellular processes. In contrast, cancer cells undergo a phenomenon termed the "Warburg effect". This is characterised by an increased glucose uptake and reliance on glycolysis for ATP production despite the availability of oxygen for oxidative phosphorylation. Most of the pyruvate is converted to lactate instead of being broken down in the mitochondria (Fig. 5.1).

For
examiner's
use

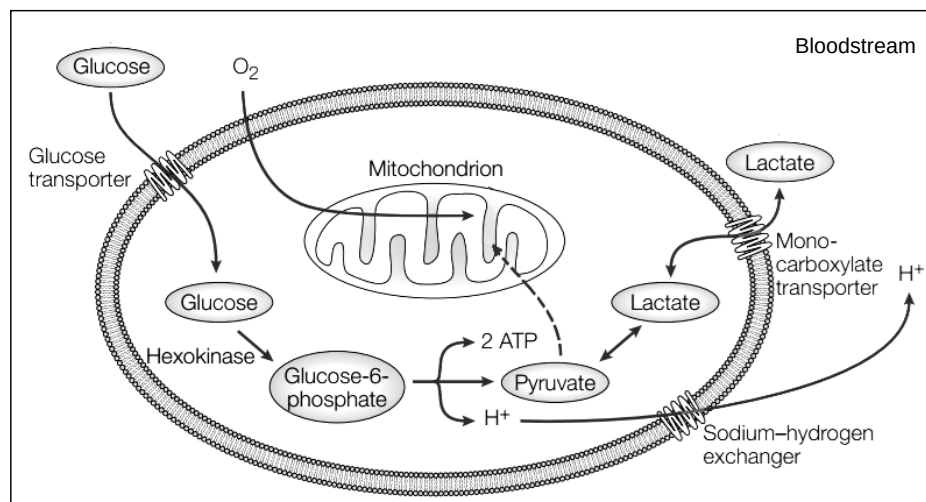


Fig. 5.1

- (a) (i) Outline the role of glycolysis in normal cells.

1. Produces net gain of 2 ATP;
2. By substrate-level phosphorylation;
3. Reduced NAD for oxidative phosphorylation;
4. Breaks down glucose into (two molecules of) pyruvate which can enter the mitochondria and is further broken down in link reaction / Krebs cycle;

- (ii) With reference to Fig 5.1 and your own knowledge, suggest why the "Warburg effect" may seem disadvantageous to the survival of the cancer cell.

1. Only net gain of 2 ATP produced as compared to 32 / many ATP under complete oxidation of glucose;
2. Consistent acidification of bloodstream due to H^+ which might result in cellular toxicity;

Note: 1 reduced NAD and 1 reduced FAD yields 2.5 and 1.5 ATP respectively.

In cancer cells, the “Warburg effect” is constitutively upregulated even under normal levels of oxygen. It is thought to be due to the reprogramming of metabolic genes to increase glucose consumption. In a research experiment, the mean glucose consumption rate of MCF-7 breast cancer cells is compared with that of the relatively more aggressive MDA-MB-231 breast cancer cells under normal and low oxygen levels (Fig. 5.2).

For
examiner's
use

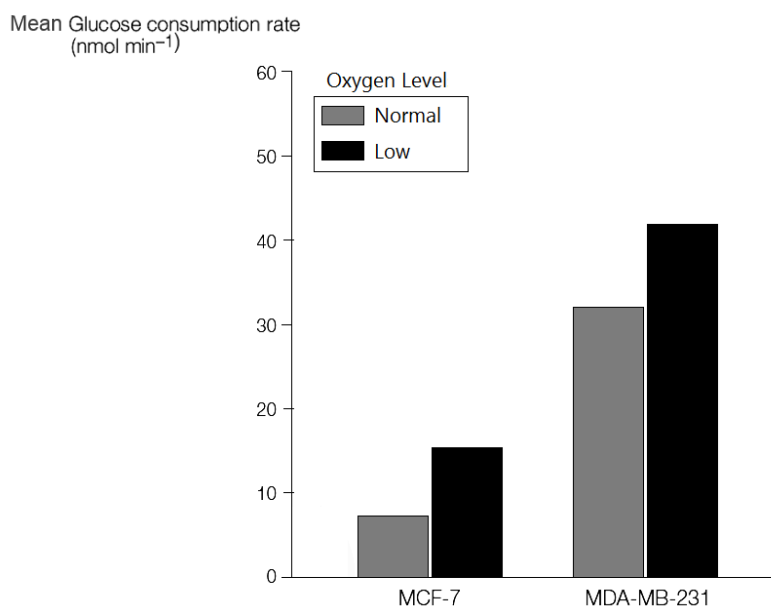


Fig. 5.2

- (b) Describe the differences in the mean glucose consumption rate between MCF-7 and MDA-MB-231 cancer cells.

1. Mean glucose consumption rate of MDA-MB-231 cells is higher than that of MCF-7 cells under both low and normal oxygen levels;
2. [Quote data] Under normal O₂ levels, MDA-MB-231 cells consume an average of 32 nmol min⁻¹ which is higher than 7 nmol min⁻¹ OR 25 nmol min⁻¹ higher;
3. [Quote data] Under low O₂ levels, MDA-MB-231 consume an average of 42 nmol min⁻¹ is higher than 15 nmol min⁻¹ OR 27 nmol min⁻¹ higher; [3]

The mTOR intracellular signalling pathway is critical for control of cell growth. Fig. 5.3 shows the signalling system that drives cell growth through greatly stimulating glucose uptake and utilisation in a normal cell.

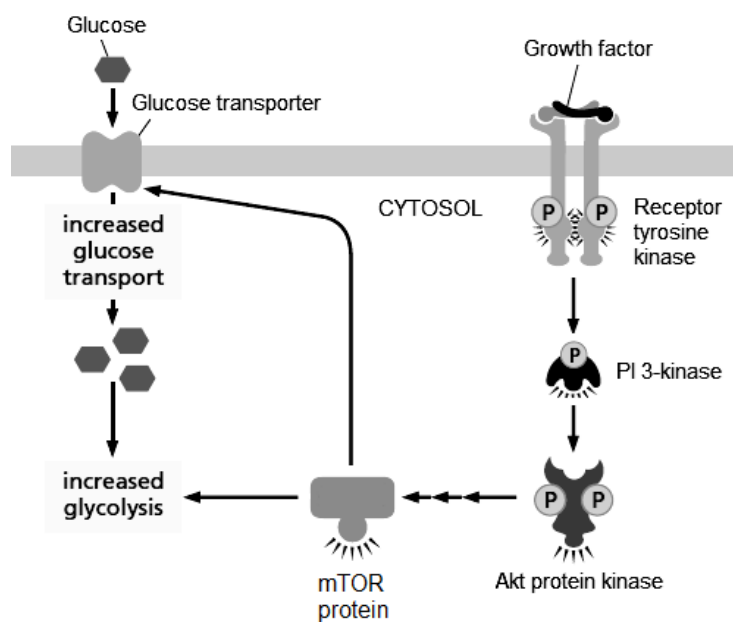


Fig. 5.3

(c) (i) Describe how the binding of the growth factor to the receptor tyrosine kinase leads to the cellular responses in Fig. 5.3.

1. Receptor tyrosine kinase dimerises and is activated;
2. then crossphosphorylates tyrosine residues on cytoplasmic tails;
3. Phosphorylates protein kinases such as PI 3-kinase / Akt protein kinase and activating them, and activated Akt protein kinase;
4. Which activates other protein kinases via a phosphorylation cascade;
5. Last protein kinase in phosphorylation cascade activates mTOR protein, increasing glucose transport / glycolysis;

[3]

(ii) Cancer cells are able to proliferate in the absence of growth factors. Suggest how the ability to do so can lead to the "Warburg effect".

Gene for receptor tyrosine kinase / PI-3 kinase / Akt / mTOR is mutated such that it is constitutively/always activated / AVP;

[1]

[Total: 11 m]

6 Fig. 6.1 shows the different stages of meiosis in a cell.

For
examiner's
use

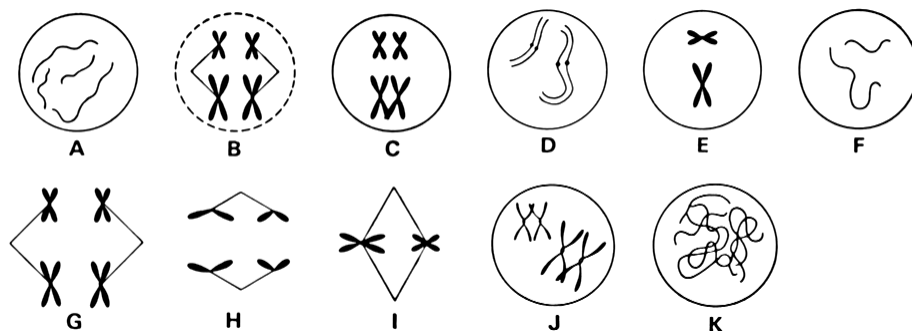


Fig. 6.1

(a) (i) Arrange the stages in chronological order. [1]

K	A	D	J	C	B	G	E	I	H	F
---	---	---	---	---	---	---	---	---	---	---

Need to get all correct to be awarded 1 m

(ii) With reference to Fig. 6.1, identify a stage which is critical for a reduction division and explain why.

1. **C – metaphase I;**
2. **Bivalents/ homologues line up along metaphase plate in 2 rows/ homologous pair line up along metaphase plate, allowing only one homologue to be present in each daughter cell at the end of meiosis I/ chromosome number is halved;**

OR

3. **B/G – anaphase I;**
4. **Separation of homologues/ homologous chromosomes results in only one homologue in each daughter cell at the end of meiosis I / chromosome number is halved;**

[2]

(iii) Explain how reduction division contributes to variation in diploid organisms.

1. **Independent assortment results in different combinations of paternal and maternal chromosomes in the gametes;**
2. **Crossing over results in different combinations of alleles in the gametes;**
3. **Random fusion of haploid gametes will increase variation;**

[2]

@1m, max 2

Some patients with the Prader-Willi syndrome have both chromosomes 15 from their mother due to an error in meiosis. Fig. 6.2 shows the possible combinations of chromosomes 15 that could be present in the gametes of the mother.

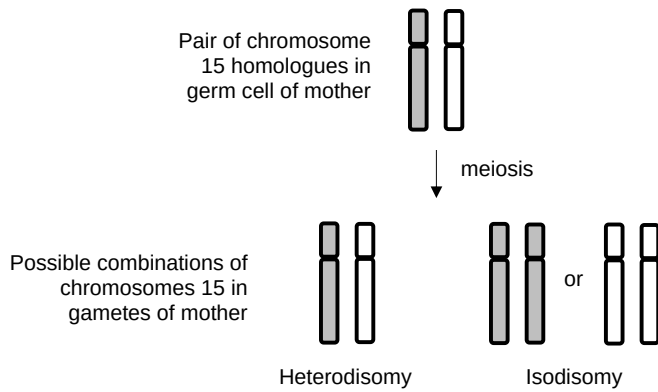


Fig. 6.2

(b) Explain how a gamete with heterodisomy of chromosome 15 could have been formed.

1. **Non-disjunction/ both homologues moved to the same pole;**
2. **In meiosis I / anaphase I of meiosis (where one daughter cell contains both homologues at the end of meiosis I and the other has no chromosome 15);**
3. **In meiosis II / anaphase II, chromatids of each homologue separate (and both daughter cells contain both homologues at the end of meiosis II);** [3]

[Total: 8 m]

- 7 The structure of a G-protein-linked receptor (GPLR) is shown from two different views in Fig. 7.1 below. Fig. 7.1a shows a GPLR from a cross section of the plasma membrane while Fig. 7.1b shows the top view of a GPLR.

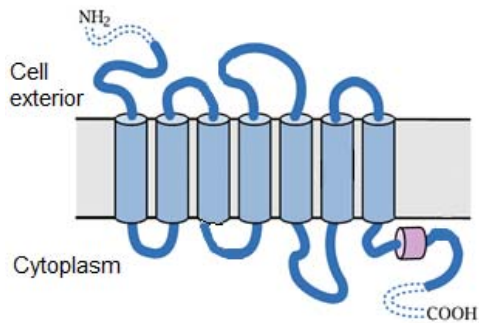


Fig 7.1a

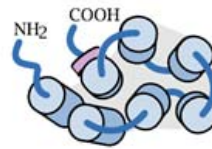


Fig 7.1b

- (a) (i) State and explain the highest level of organization of the GPLR.

1. **Tertiary structure;**
2. **A single polypeptide chain (shown by one NH₂ COOH group) is folded into a globular structure / 3D conformation.**

[2]

- (ii) Identify a hydrophilic domain of the GPLR and describe its corresponding function.

1. **The binding site for signal molecule (facing exterior of the plasma membrane)**
2. **This allows signal molecule to bind and trigger conformational change in GPLR, activating it.**
3. **The binding site for G protein (facing cytoplasmic face of the plasma membrane).**
4. **Activated GPLR binds to G protein and activates G protein when GTP displaces GDP.**

* Points 1 & 2, or points 3 & 4

[2]

For
examiner's
use

Recent studies have found that GPLRs have alternative binding sites other than the ligand-binding sites. Allosteric molecules bind to these alternative binding sites to change cellular response levels. One such molecule has been found to bind to the GPLR specific to glucagon.

One of the normal cellular responses of glucagon binding is the inhibition of glycogenesis. The rate of glycogenesis with and without the allosteric molecule is shown in Fig. 7.2.

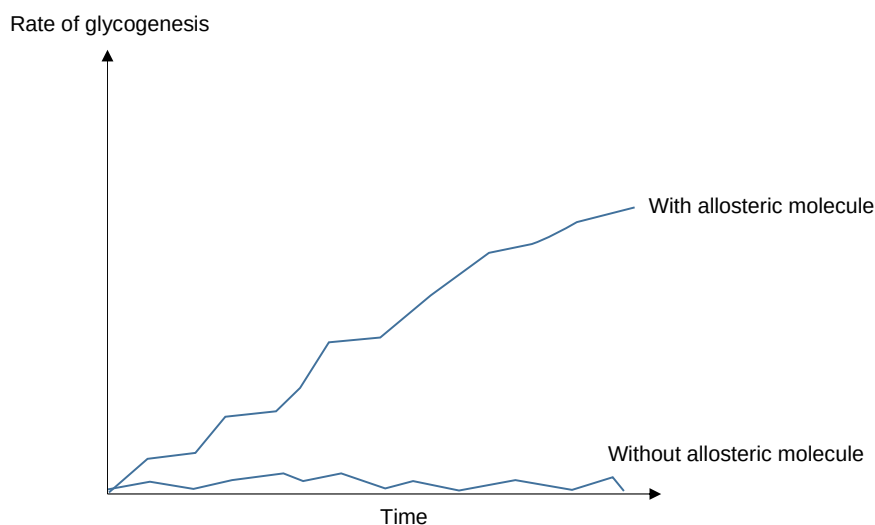


Fig. 7.2

- (b) (i) Based on the information above, state if the allosteric molecule is an activator or an inhibitor of GPLR.

Inhibitor; [1]

- (ii) Explain the effect of the allosteric molecule on the rate of glycogenesis.

1. With the binding of the allosteric inhibitor, the rate of glycogenesis steadily increases;
 2. When the allosteric inhibitor binds, 3D conformation of GPLR changes;
 3. G protein (is no longer complementary to and) will not be able to bind to and activate adenylyl cyclase;
 4. Adenylyl cyclase will not produce cAMP, the second messenger;
 5. cAMP will not be able to bind and activate Protein Kinase A and the phosphorylation cascade will not occur;
 6. Glycogen phosphorylase will not be activated, thus glycogenolysis will not occur;
- [5]

[Total: 10 m]

- 8 Speciation events have been observed to occur very frequently in bacteria. It was suggested that the high rate of speciation is due to the high level of variation in bacteria.

(a) Transformation and conjugation are two processes which increase the level of variation in bacteria.

Distinguish these two processes.

	Conjugation	Transformation
Genetic material being transferred	1. Transfers the F plasmid or R plasmid	1. Transfers fragments of the bacterial chromosome;
Donor cell	2. Donor cells are F ⁺ cells / carries the F plasmid or R plasmid;	2. Donor cells may be lysed cells that releases its DNA;
Recipient cell	3. Recipient cells are F ⁻ cells / do not carry the F plasmid or R plasmid;	3. Recipient cells must be competent / secrete the competence factor;
Physical contact	4. Direct physical contact required between two cells through sex pilus;	4. No physical contact required between cells / exogenous DNA taken up directly by recipient cell;

[3]

Bacterial evolution is one of the most dynamic and exciting areas in current biological research.

Over the years, a barrier in this field of research is the difficulty in classifying bacterial species. However, in recent times, new analytical tools in molecular biology have offered new insights into the classification of bacterial species.

(b) (i) Suggest why scientists had difficulties in the classification of bacterial species.

1. Bacteria reproduce asexually / by binary fission;
2. Unable to determine between species according to biological species concept / bacteria are unable to interbreed to produce fertile viable offspring;
3. Horizontal gene transfer/transformation/conjugation between relatively distantly related bacteria / different bacterial species (OWTTE);
4. Results in high rates of recombination, making it difficult to determine a single species;
5. Different species may be morphologically similar;

[2]

(ii) Explain how analytical molecular tools have helped overcome this barrier in research.

1. Molecular tools have helped determine genetic sequences of bacteria / compare genetic sequences of bacteria;
2. Allowing scientists to classify bacteria according to genetic distance / phylogenetic distance;
3. Provides an objective method (to determine genetic distance);
4. Data obtained is quantitative;
5. Hence more sensitive to differences between species / data could be used for statistical analyses;

[2]

- (c) The outer layers of the two types of bacteria with peptidoglycan cell walls known as Gram-positive and Gram-negative bacteria are shown in Fig. 8.1 below.

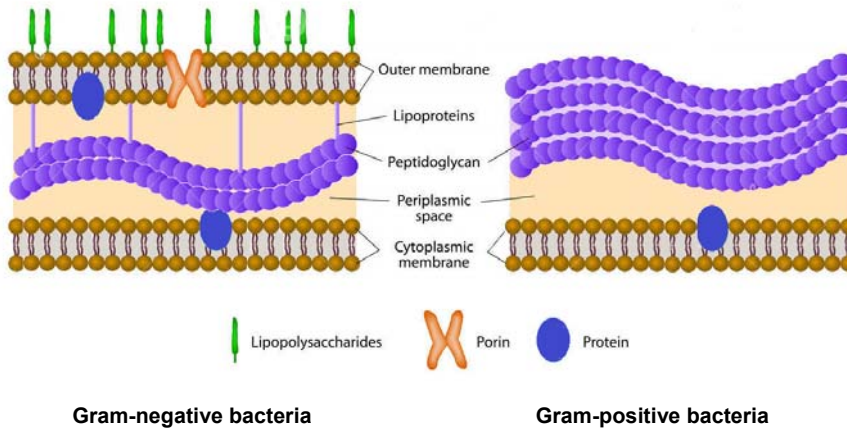


Fig. 8.1

Penicillin is an antibiotic that is known to be effective against only one of the two types of bacteria above.

With reference to the information given and your own knowledge, deduce which type of bacteria is susceptible to the action of penicillin and explain why.

1. Penicillin is effective only against gram-positive bacteria;
2. Gram-negative bacteria have an outer membrane (and lipoproteins) (that surrounds the peptidoglycan layer of the cell wall) / penicillin can directly access the peptidoglycan cell wall in gram-positive bacteria;
3. Preventing the action of penicillin as penicillin inhibits the crosslinks in peptidoglycan cell wall;

[3]

- (d) Explain how antibiotic-resistant bacteria can become increasingly common in a population of bacteria.

1. Horizontal gene transfer (transformation, transduction, conjugation) occurs which increases genetic variation in the bacteria;
2. Genetic variation exists in the form of antibiotic sensitivity and antibiotic resistance;
3. Antibiotics act as selection pressure;
4. Non-resistant bacteria are selected against / bacteria which are resistant to antibiotics are selected for / they have a selective advantage;
5. Allele coding for antibiotic resistance passed down to subsequent generations of bacterial cells (during binary fission);
6. Over many generations, frequency of allele coding for antibiotic resistance increases in the gene pool;

[4]

[Total: 14 m]

- 9 Cyanobacteria are aquatic blue-green bacteria which are highly similar to algae as they can obtain their energy through photosynthesis. They are typically found in tropical waters due to their ability to thrive in bright and warm areas. Generally found on the surface of lakes and oceans, they can reproduce exponentially and cause a rapid increase in their population known as blooms. Certain genera of blooming cyanobacteria such as *Microcystis* can produce cyanotoxins which, in high concentrations, can poison and even kill animals and humans.

A study was conducted to find out the effects of temperature on the maximum growth of *Microcystis aeruginosa* as well as three other harmless green algal species (P, Q and R) which are a main source of food for aquatic animals (Fig. 9.1). The global mean sea surface temperature anomalies, which indicate differences in temperature when compared to the baseline temperature in 1880 were also recorded (Fig. 9.2).

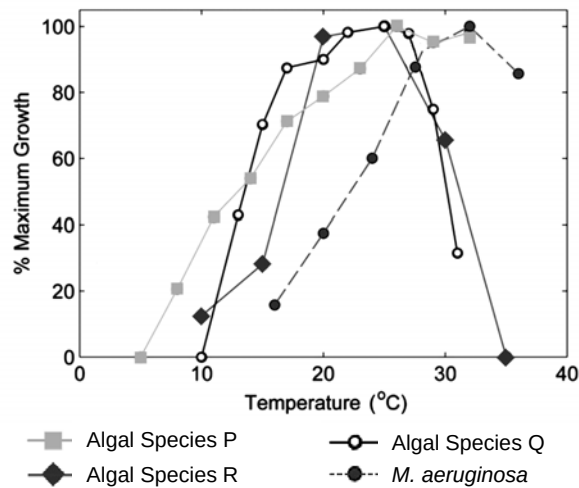


Fig. 9.1

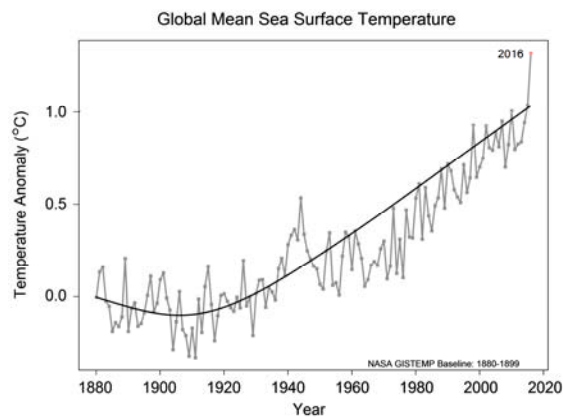


Fig. 9.2

(a) Describe the effect of temperature on the maximum growth of *M. aeruginosa*.

1. Maximum growth increases from 18% to 100% from 16 to 33°C;
2. And decreases from 100% to 84% from 33 to 37°C; [2]

(b) Explain how human activities could have contributed to an increase in sea surface temperatures.

1. Burning of fossil fuels such as coal, natural gas and oil releases greenhouse gases such as CO₂ and CH₄, causing global temperatures to rise;
2. Deforestation leads to diminishing carbon sink, and burning of trees and soil disturbance releases CO₂;
3. Due to increased meat consumption, more livestock is reared which releases CH₄ due to enteric fermentation;
4. Hence increases in global temperatures will lead to increase in sea surface temperatures; [3]

(c) With reference to the information provided, discuss the impact of *M. aeruginosa* on the global food supply of humans in the future.

1. May adversely affect food supply;
2. Fig. 9.1 shows increasing trend of maximum growth of *M. aeruginosa* with increasing temperatures before 33°C;
3. Where SST is increasing as shown in Fig. 9.2 by an increase of 1.4°C / 1.0°C from 1880 to 2016;
4. Cyanotoxins produced by blooming *M. aeruginosa* harm fishes, hence affecting fisheries and fish supply for humans;
5. Cyanobacterial distribution will expand polewards and affect more fisheries in temperate areas as well;
6. Blooms of *M. aeruginosa* may also act as competition with other algal species which may be a food source for other fish;
7. May not adversely affect food supply;
8. Fig. 9.1 shows decreasing maximum growth of *M. aeruginosa* beyond 33°C;
9. Due to denaturation of enzymes and hence metabolic rate;
10. As *M. aeruginosa* is unable to survive at higher temperatures; [5]

[Total: 10 m]

For
examiner's

Commented [LKM1]: Students must be able to identify different parts of the graph and explain them, while identifying trends (increasing trend)

Commented [LKM2]: Direct recall from LO – students have to phrase their answers in the context of SST

Commented [LKM3]: Similar to CA2 Q phrasing, where student can try to provide evidence where global food supply may be increased or decreased / assess validity of information in making such a claim